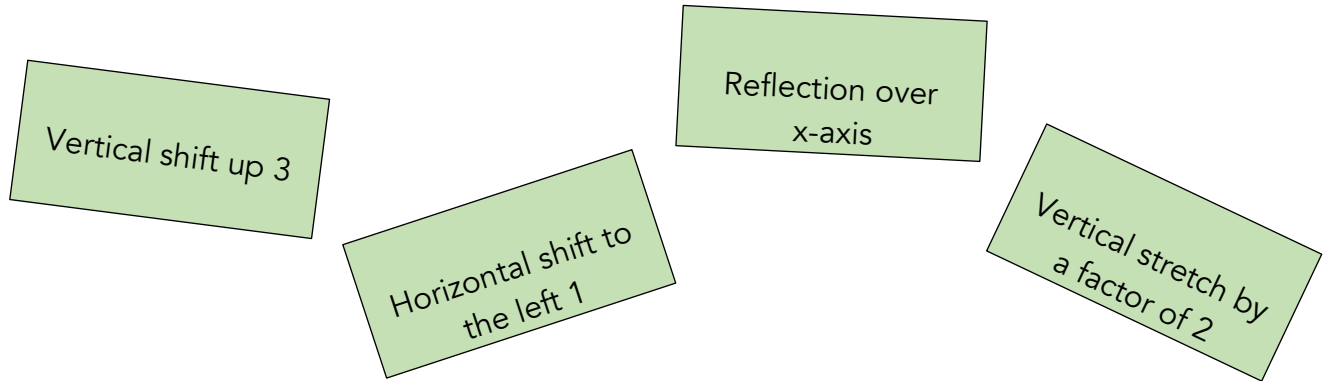


Transformation Mix-Up

I took the graph $y = 3x^2 - 5$ and performed the four transformations shown on the cards below.



Unfortunately, I can't remember the order in which I carried out the four transformations. All I know is that I ended up with the graph of $y = -6(x + 1)^2 + 7$.

Can you find an order in which I could have carried out the transformations? Is there more than one way of doing this? If so, can you find them all?

vertical stretch, horizontal shift, vertical shift, reflection

yes, there are multiple ways because the horizontal shift could occur in any of the 4 positions without changing the final result.

There are 4 possibilities.

Can you explain why different orders can lead to the same outcome?

The horizontal shift is the only transformation affecting the independent variable and thus is placed in the parentheses and is not affected by stretches, reflections over x-axis or vertical shifts.

What other parabolas could I have ended up with if I had performed the four transformations in a different order?

$-6(x+1)^2 + 4$ v. shift, v. stretch, reflection OR
v. shift, reflection, v. stretch

$-6(x+1)^2 + 16$ reflection, v. shift, v. stretch

$-6(x+1)^2 + 13$ reflection, v. stretch, v. shift

v. stretch, reflection, v. shift